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# Numerical Investigation of Fuel Property Effects on Mixed-Mode Combustion in a Spark-Ignition Engine

*In this study, lean mixed-mode combustion is numerically investigated using computational fluid dynamics (CFD) in a spark-ignition engine. A new E30 fuel surrogate is developed using a neural network model with matched octane numbers. A skeletal mechanism is also developed by automated mechanism reduction and by incorporating a $NO_x$ submechanism. A hybrid approach that couples the G-equation model and the well-stirred reactor model is employed for turbulent combustion modeling. The developed CFD model is shown to well predict pressure and apparent heat release rate (AHRR) traces compared with experiment. Two types of combustion cycles (deflagration-only and mixed-mode cycles) are observed. The mixed-mode cycles feature early flame propagation and subsequent end-gas auto-ignition, leading to two distinctive AHRR peaks. The validated CFD model is then employed to investigate the effects of $NO_x$ chemistry. The $NO_x$ chemistry is found to promote auto-ignition through the residual gas, while the deflagration phase remains largely unaffected. Sensitivity analysis is finally performed to understand effects of fuel properties, including heat of vaporization (HoV) and laminar flame speed ($S_L$). An increased HoV tends to suppress auto-ignition through charge cooling, while the impact of HoV on flame propagation is insignificant. In contrast, an increased $S_L$ is found to significantly promote both flame propagation and end-gas auto-ignition. The promoting effect of $S_L$ on auto-ignition is not a direct chemical effect; it is rather caused by an advancement of the combustion phasing, which increases compression heating of the end-gas.* [DOI: 10.1115/1.4048242]

*Keywords: mixed-mode combustion, spark-ignition, detailed chemical kinetics, fuel property sensitivity*

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## Introduction

Lean combustion is beneficial to spark-ignition (SI) engine operation due to the higher efficiency and lower emissions than conventional stoichiometric engine operation. However, applications of lean combustion are challenged by the intrinsically low flame speeds and the susceptibility to static and dynamic instabilities.

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To overcome these difficulties, spark-assisted compression ignition (SACI) or mixed-mode combustion is a promising strategy, which combines conventional deflagrative flame propagation and controlled end-gas auto-ignition. Thus, the fuel can burn sufficiently fast through auto-ignition, compensating for the low flame speed of lean mixtures, while the engine remains knock free.

Lean combustion under mixed-mode conditions has been extensively studied by experiments [1–4]. Urushihara et al. [1] studied spark-ignited compression ignition and demonstrated the increased engine load compared to conventional homogeneous charge compression ignition (HCCI) combustion. Zigler et al. [2] studied SACI in an optical engine and identified the presence of spark-initialized turbulent flame propagation and subsequent auto-ignition in the end-gas. Sensitivity of the air preheating and spark timing on various engine performance metrics was also investigated with high-speed imaging. Ma et al. [5] used $CH_2O$ and OH chemiluminescence to investigate in-cylinder combustion behaviors in flame-induced auto-ignition. A stoichiometric condition was employed, while the observed flame characteristics—occurrence of auto-ignition in the outer rim of the deflagrative flame front and accelerated reaction front propagation—could hold true for lean engine operations as well. Sjöberg and Zeng [3] studied mixed-mode combustion at lean and diluted conditions with various fuels. Significant cycle-to-cycle variability (CCV) was observed at ultra-lean conditions, which could pose a great challenge for practical engine operation. Reuss et al. [6] demonstrated that the early kernel growth was a major source for CCV under SACI conditions. Hu et al. [4] identified injection strategies that could stabilize the ultra-lean operation and improve combustion efficiency for mixed-mode combustion.

In this context, computational fluid dynamics (CFD) modeling has its unique capabilities to probe into the governing physics of mixed-mode combustion, providing unique opportunities for studying CCV and fuel property effects. Dahms et al. [7] developed a mixed-mode flamelet combustion model, which combines the SparkCIMM ignition model, the G-equation model and multi-zone chemistry, targeting spark-assisted HCCI engines at lean conditions with significant exhaust gas recirculation dilution. The model demonstrated a good agreement between CFD and experiment. However, the target operation was at a relatively low-CCV condition, and the performance under high-CCV conditions remains unclear. Middleton et al. [8] studied SACI combustion at stoichiometric conditions and investigated the effect of spark timing and charge temperature on combustion phasing and heat release rate that are governed by the competition between flame propagation and auto-ignition, using the coherent flamelet model coupled with detailed chemistry. However, few CFD studies till date have focused on mixed-mode combustion under lean and dilute conditions with a high level of CCV.

The objectives of this study are, therefore, twofold. The first goal is to develop an engine CFD model that can accurately capture lean, mixed-mode combustion characteristics. The second objective is to identify effects of chemical and physical fuel properties on mixed-mode engine performance, which can eventually enable co-optimization of fuels and engines.

## Engine Specifications and Operating Conditions

The engine simulated in this study is a single-cylinder, four-valve, direct-injection spark-ignition (DISI) research engine at Sandia National Laboratories. Figure 1 schematically shows the cross section of the combustion chamber at the top dead center (TDC). A long-reach spark plug is adopted to extend the spark plasma to the center of the combustion chamber, which can potentially improve the ignition efficiency, especially for lean operations. The fuel injector is mounted on the pent-roof facing the spark plug allowing for direct injection of fuel into the chamber center. The piston bowl window can provide optical access to the combustion chamber, but in this study, a metal blank was used to enable continuously fired all-metal engine experiments. One of the intake valves was deactivated to enhance the in-cylinder swirl level and thereby the overall mixing process. Relevant engine specifications are provided in Table 1.

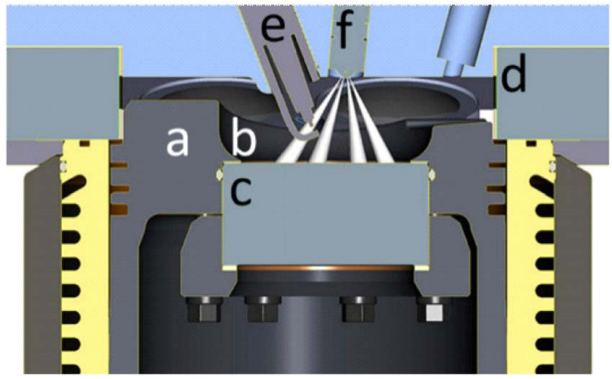



**Fig. 1 Schematic of Sandia DISI engine [3] showing the cross section of the combustion chamber at TDC: a, piston; b, piston bowl; c, piston bowl window; d, pent-roof window; e, spark plug; and f, fuel injector**

The engine is operated at a lean condition (a global fuel/air equivalence ratio of 0.55) using certification gasoline blended with 30% ethanol by volume (referred to as "E30" hereinafter). To achieve a well-mixed charge of fuel, air, and residual gas (~6%), the fuel was injected using three injections of equal duration during the intake stroke. Research octane number (RON) and motor octane number (MON) of the E30 fuel are 105 and 91, respectively. To achieve mixed-mode combustion with maximum brake torque for such a high-octane fuel, a fairly advanced combustion phasing is necessary, requiring the use of an advanced spark timing (−57 CA ATDC). However, such an early spark timing leads to a significant level of CCV. Capturing CCV for this operating point will be one focus of the present modeling efforts. Engine operation conditions are summarized in Table 2, and more details on engine configuration and operating conditions are presented in Ref. [3].

## Numerical Approach

**Computational Fluid Dynamics Geometry and Model Setup.** To model the Sandia DISI engine, a full-scale engine geometry, including intake and exhaust runners, the piston head, the

**Table 1 Engine specifications**

| Parameter | Value |
|---|---|
| Bore | 86.0 mm |
| Stroke | 95.1 mm |
| Connecting rod length | 166.7 mm |
| Piston pin offset | −1.55 mm |
| Compression ratio | 12:1 |

**Table 2 Engine operating conditions**

| Parameter | Value |
|---|---|
| Engine speed | 1000 rpm |
| IMEPg | 446 kPa |
| Intake temperature | 100 °C |
| Intake pressure | 87.0 kPa |
| Exhaust pressure | 100.1 kPa |
| Injection timings | −318, −303, −288 CA ATDC |
| Injection duration | 444 $\mu$s |
| Injected fuel mass | 17.8 mg/cycle |
| Spark timing | −57 CA ATDC |



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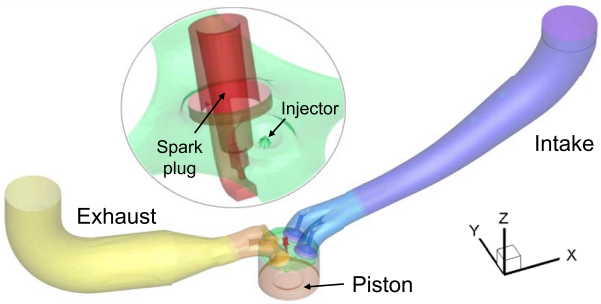



**Fig. 2 Engine geometry used in simulations**

piston, the spark plug, and the fuel injector, is used, as shown in Fig. 2. The engine is simulated using the CONVERGE code v2.4 [9]. The re-normalized group $k-\epsilon$ model is used to describe the Favre-averaged turbulent flow. Wall heat transfer is modeled with a temperature wall function from Amsden and Findley [10]. Cylinder wall temperature is set to be 445 K. Experimentally measured high-speed intake and exhaust pressures (varying with time) are specified at the intake port inlet and the exhaust port outlet. Fuel spray and in-cylinder combustion processes are simulated with the Eulerian–Lagrangian approach. The spray injection is described by the blob injection approach [11], while droplet breakup, droplet evaporation, and drag force are modeled using the Kelvin–Helmholtz and Rayleigh–Taylor models [12,13], the Frossling correlation [14], and a dynamic drag model [15], respectively. Liquid properties are taken from a previous study [16] on the same engine platform with the same E30 fuel.

For turbulent combustion modeling, a hybrid approach is employed to capture mixed-mode combustion. In particular, the G-equation model is employed to track deflagrative flame propagation with tabulated laminar flame speed. A passive scalar $G$ is transported according to the instantaneous turbulent flame speed, which is modeled using Peter's model [17]. The value of $G$ indicates the distance from a local fluid element to the mean flame front. $G=0$ identifies the flame front location, while $G<0$ and $G>0$ indicate the unburned and burned mixtures, respectively. The laminar flame speed is calculated based on one-dimensional (1D) freely propagating premixed flames and is then tabulated as a function of pressure, unburned temperature, local equivalence ratio, and local dilution ratio. The local dilution ratio is calculated through a separate passive transport equation. The well-mixed model coupled with detailed chemical kinetics is used to predict auto-ignition in the end-gas. The multi-zone model is further employed to accelerate detailed chemistry integration. This hybrid approach has been demonstrated to be able to capture knock in Cooperative Fuel Research engines [18,19] and boosted SI engines [20]. A unique feature of this hybrid approach is that it allows isolated investigation of individual chemical properties such as flame speed and ignition delay.

A modified cut-cell Cartesian grid method for automatic mesh generation is used during runtime [9]. The base grid size is $\Delta_0=4$ mm, and the minimum grid size is $\Delta_5=0.125$ mm. In particular, fixed embedding is applied to better resolve in-cylinder dynamics ($\Delta_2=1$ mm), wall boundary layers ($\Delta_3=0.5$ mm), spray injection ($\Delta_4=0.25$ mm), early flame propagation ($\Delta_5=0.125$ mm), and other small geometrical structures ($\Delta_3=0.25$ mm). Adaptive mesh refinement based on velocity and temperature fluctuations is further employed to better resolve complex flow and flame structures with a minimum cell size of $\Delta_3=0.5$ mm. Note that in $\Delta_n$, $n$ represents the level of mesh refinement with respect to the base grid size. The peak cell count during a full engine cycle is approximately 1.6 million. The computational cost for simulation of one engine cycle is approximately two days.

**E30 Fuel Surrogate.** To generate the surrogate composition for gas-phase modeling, a nonlinear regression model was employed

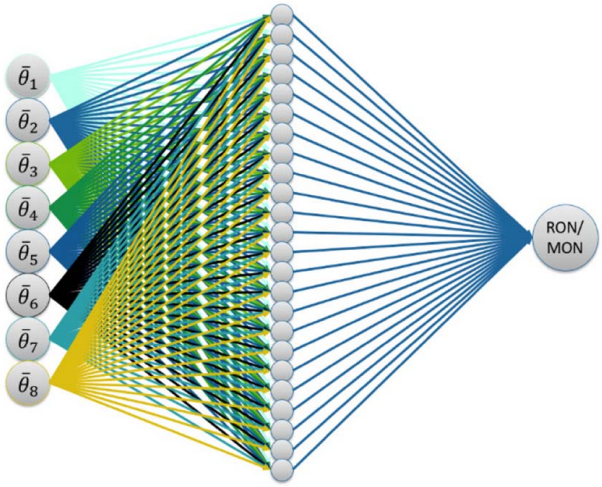



**Fig. 3 A schematic of the neural network architecture [30] used for fuel surrogate development**

**Table 3 E30 fuel surrogate composition (by mole)**

| *n*-Heptane | *iso*-Octane | Toluene | Ethanol |
|---|---|---|---|
| 6.0% | 16.8% | 28.4% | 48.8% |

that could relate the ignition chemistry from a detailed chemical kinetics model [21] and other thermophysical properties to RON and MON. In this case, the nonlinear regression model was a feed-forward neural network [22]. The regression was an approximation, but it could balance the error in the chemical kinetic model with the error correlating octane numbers to ignition delay times [23–28]. The model could be evaluated in less than 10 s on a single CPU thread for compositions containing any combination of the more than 50 hydrocarbons and biofuels represented in the detailed chemical kinetics model developed by Mehl et al. [21]. This model was then combined with standard optimization routines [29] to find the fuel blend with equivalent octane ratings within the accuracy of the regression.

Two feed-forward neural networks were created, one for each octane number. They both used the same inputs and architecture—a single hidden layer with 24 nodes. The inputs included three ignition delay-related quantities, simulated using the detailed chemistry model [21] in a homogeneous, constant-volume reactor at 825 K and 20 bar. These were the inverse of the ignition delay time to reach 1225 K and the derivative of the normalized ignition delay time with respect to pressure and temperature. The other neural network inputs were the enthalpy of vaporization and liquid density at 298 K, and the mole-averaged atom counts for hydrogen, carbon, and oxygen. A schematic of the neural network architecture is shown in Fig. 3. The neural network was demonstrated to have good predictive capability with root-mean-square errors in RON/MON of approximately 1 ON for the cross-validation data. Further details on the design, implementation, and validation of the neural network can be found in Ref. [30].

A fuel surrogate based on toluene primary reference fuel (TPRF) and ethanol was obtained using the basic multivariable minimization techniques found in the PYTHON SCIPY library[2] in conjunction with the neural network regression models. Specifically, the difference between predicted and target RON (105) and MON (91) with respect to component volume fractions was minimized, subject to the constraints of 30% ethanol volume fraction and the sum of

[2]https://www.scipy.org/



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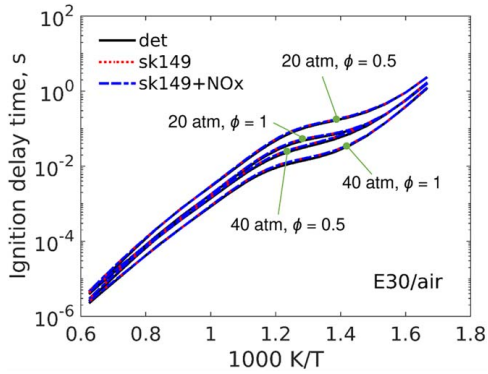



**Fig. 4 A comparison of ignition delays calculated using the detailed model ("det"), the skeletal model without $NO_x$ ("sk149"), and the final skeletal model with $NO_x$ ("sk149 + $NO_x$")**

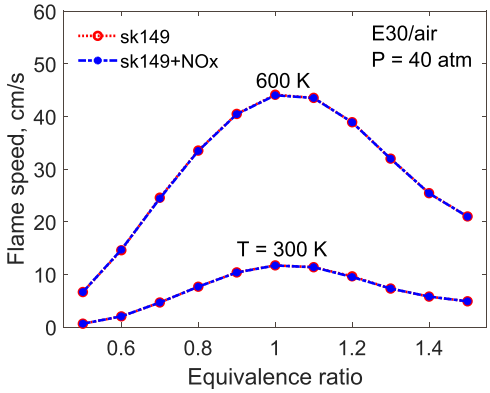



**Fig. 5 A comparison of laminar flame speeds calculated using the skeletal model without $NO_x$ ("sk149") and the final skeletal model with $NO_x$ ("sk149 + $NO_x$")**

the volume fractions being unity. The resultant TPRF–ethanol surrogate is presented in Table 3.

**E30 Skeletal Reaction Model.** The detailed chemical kinetic model [21] of the proposed TPRF–ethanol blend consists of 2878 species and 12,839 reactions, which is prohibitive for three-dimensional (3D) engine CFD simulations. Therefore, mechanism reduction based on directed relation graph [31] and sensitivity analysis is employed to systematically reduce the size of the reaction model. The reduction is performed based on a large set of reaction states sampled over the parameter range of pressure from 1 to 100 atm, equivalence ratio from 0.3 to 2.0, inlet temperature of 300 K for perfectly stirred reactors, and initial temperature from 600 to 1600 K for auto-ignition, covering the low-temperature chemistry region that is important for engine combustion. The error tolerance used in the reduction is 0.3, implying that the worst case error of the skeletal mechanism is 30%. The resultant skeletal model consists of 149 species and 640 reactions. A submechanism of $NO_x$ chemistry[3] is then merged into the skeletal model, resulting in a final rection model containing 164 species and 694 reactions.

Figure 4 compares the ignition delays of the final skeletal model with $NO_x$ against the skeletal model without $NO_x$ as well as the detailed model at different temperatures and pressures. Excellent agreement is observed between the skeletal models and the detailed model for both ignition delay and flame speed. Laminar flame speeds calculated by the two skeletal mechanisms are also compared in Fig. 5. The addition of $NO_x$ has negligible impact on 0D and 1D calculations at the selected conditions. However, $NO_x$ chemistry can be important under practical engine conditions due to the presence of the residual gas, as will be investigated in Results and Discussion section.

[3]http://combustion.ucsd.edu

**Table 4 Predicted and measured mean combustion characteristics**

| Quantity | $P_{max}$ (MPa) | IMEPg (MPa) | CA10 | CA50 | CA90 |
|---|---|---|---|---|---|
| Experiment | 3.93 | 0.446 | –8.04 | 3.54 | 22.3 |
| CFD | 4.08 | 0.497 | –14.1 | 2.22 | 21.4 |

## Results and Discussion

**Model Performance.** The proposed modeling approach is first validated in this section. Table 4 presents the comparison of key engine performance parameters, including peak cylinder pressure ($P_{max}$), gross indicated mean effective pressure (IMEPg), CA10, CA50, and CA90, obtained from simulations and experimental measurements. Predicted values overall agree well with measured ones. A slightly earlier combustion phasing (CA10 and CA50) predicted by simulation is possibly due to the use of a simplified ignition model (a spherical energy source at the center of the spark gap) during the energizing stage. However, the computational cost is significantly reduced with this simplified ignition model.

Figure 6 shows the pressure and AHRR traces obtained from the experiment (500 cycles) and the simulation (13 cycles). Good agreement is observed between the simulation and experimental data, with the predicted mean pressure being slightly higher than the measured mean pressure. In addition, the moderate level of, but not full range of, CCV is captured by CFD. This is because unsteady Reynolds-averaged Navier–Stokes (RANS) models solve time-averaged Navier–Stokes equations and therefore intrinsically

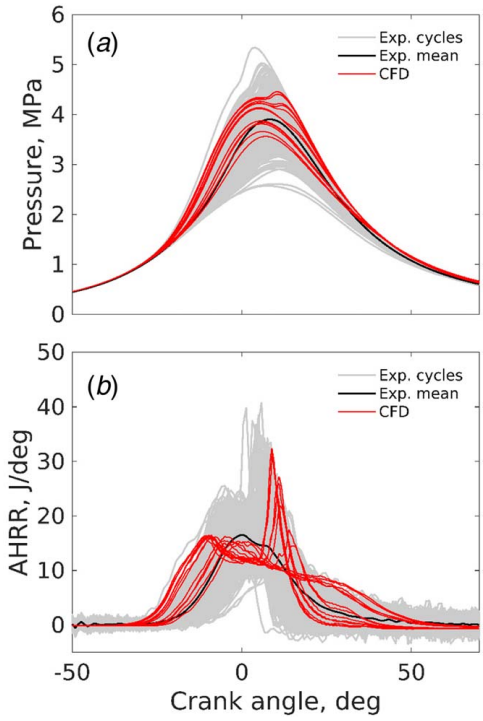



**Fig. 6 Predicted (*a*) pressure and (*b*) apparent heat release rate (AHRR) profiles compared with experimental values. Individual experimental cycles, mean experimental values, and numerical cycles are shown in gray, black, and red, respectively. (Color version online.)**

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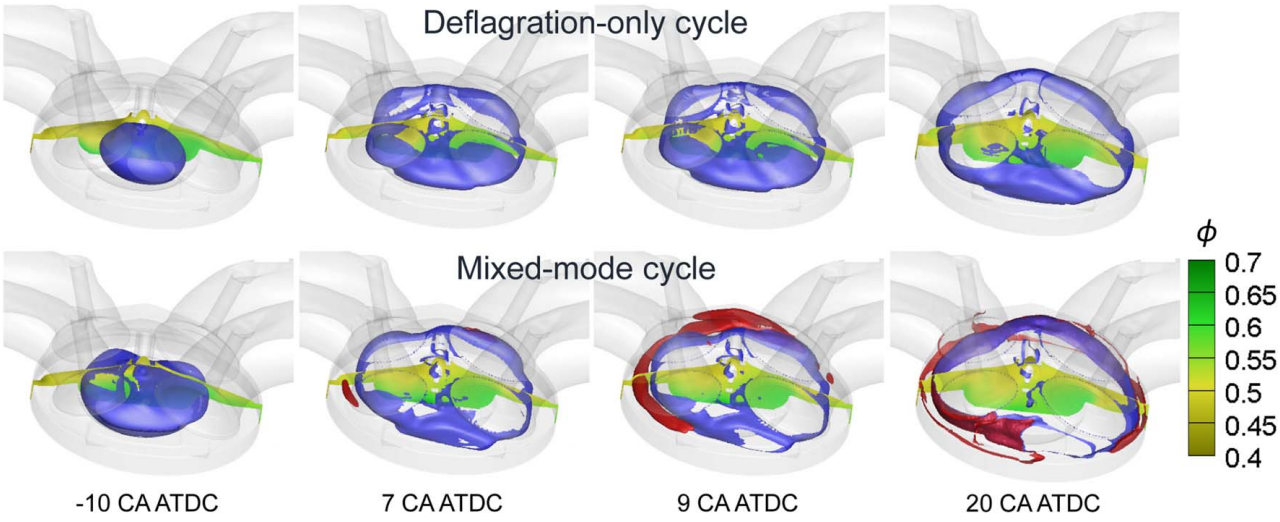



**Fig. 7 Evolution of deflagrative flame fronts (blue) and auto-ignition fronts (red) in a selected deflagration-only (top) and a selected mixed-mode cycle (bottom), along with the distribution of equivalence ratio ($\phi$) on a $x-z$ plane cutting through the center of the spark gap. Blue and red isosurfaces represent deflagrative and auto-ignitive fronts, respectively. Deflagrative fronts are identified by $G=0$, while auto-ignitive fronts are identified by $Y_{H_2O}/Y^{eq}_{H_2O}$=70% in the end-gas, i.e., the region where $G<0$. (Color version online.)**

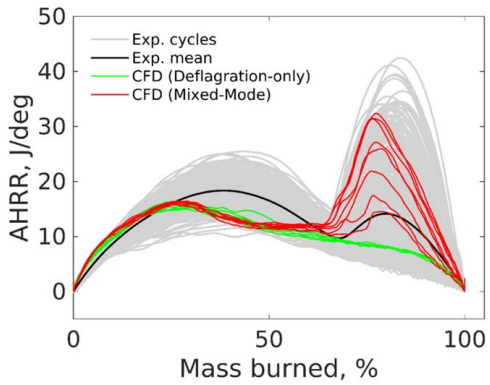



**Fig. 8 Apparent heat release rate versus burned mass fraction, obtained from experiment and simulation**

predict lower CCV. Two types of combustion cycles are observed in both experiment and simulation (Fig. 6(*b*)). The first type of cycles features low in-cylinder pressure and heat release rate, resulting in a single AHRR peak. This type of combustion cycles is similar to those observed in conventional SI engines (although the combustion duration is typically longer due to the lean condition) and is referred to as deflagration-only cycles. The other type of cycles shows higher in-cylinder pressure and heat release rate and exhibits two AHRR peaks. The first and second peaks correspond to the early flame propagation and the subsequent end-gas auto-ignition processes, respectively. This type of cycles is, therefore, referred to as mixed-mode cycles. Figure 7 shows the flame structure and dynamics of the two types of combustion cycles, namely deflagration-only (top) and mixed-mode cycles (bottom). In contrast to the deflagration-only cycle, earlier flame propagation is seen for the mixed-mode cycle, and isolated ignition spots are formed (~7 CA) followed by volumetric auto-ignition in the end-gas. As end-gas auto-ignition rapidly consumes the reactants ahead of the flame fronts (7–20 CA), turbulent flame propagation due to deflagration is still present, although much slower than auto-ignition.

The two types of combustion cycles can further be distinguished from each other in the mass burned space as shown in Fig. 8, where burned mass fraction is calculated as the integrated heat release rate normalized by total heat released from each cycle. It is further seen that the initial flame propagation phase in the two types of combustion cycles are very similar to each other, while the mixed-mode cycles feature a second peak at ~75% mass fraction burned (~80% in experiment). The presence of the second peak is, therefore, employed as a criterion to systematically distinguish between these two types of cycles, without specifying any empirical threshold. With this criterion, the predicted fraction of mixed-mode cycles from the simulation is 61.5%, closely matching the experimental value 63.2%. Mixed-mode combustion cycles are further characterized by the mean formaldehyde mass fraction ($Y_{CH_2O}$) inside the cylinder, versus the burned mass fraction, as shown in Fig. 9. Both types of cycles exhibit an initial plateau, indicating stable flame propagation. Compared with deflagration-only cycles, mixed-mode combustion cycles feature a rapid increase in $Y_{CH_2O}$ near CA50, which leads to fast auto-ignition. The observed difference in evolution of $Y_{CH_2O}$ can be explained as follows. In the flame propagation mode, $CH_2O$ is produced only within a thin layer (the preheat zone) ahead of the flame fronts, and therefore, $Y_{CH_2O}$ is closely related to the total flame surface area, which does not vary significantly during a large part of the main heat release process. When chemical reactions in the end-gas are nonnegligible, the low-to-intermediate temperature chemistry starts to build up radical pools in the fresh mixture, and thus, $Y_{CH_2O}$ increases exponentially until volumetric auto-ignition consumes it.

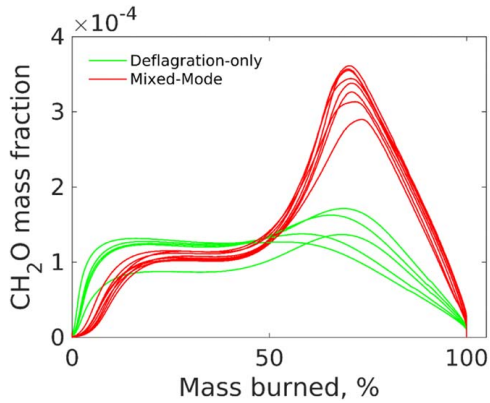



**Fig. 9 Formaldehyde mass fraction versus burned mass fraction, obtained from simulation**

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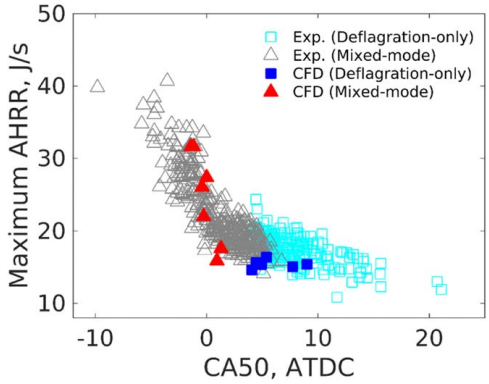



**Fig. 10 Peak heat release versus CA50 based on experimental and numerical results**

The difference between mixed-mode and deflagration-only cycles, and their correlations with combustion phasing are further investigated. Figure 10 shows the scatter of CA50 as function of peak heat release rate for all the simulation cycles overlaid on the experimental data. It is clear from both experimental and simulation results that mixed-mode combustion occurs with more advanced CA50. This is mainly because earlier flame propagation promotes auto-ignition by increasing in-cylinder pressure and temperature. The well-predicted correlation between mixed-mode combustion tendency and CA50 therefore further demonstrates the accuracy of the developed CFD model.

**Effects of $NO_x$ Chemistry.** While the overall lean operation would generally reduce the production of thermal NO, the high octane number of the current E30 fuel forces the use of a fairly advanced combustion phasing to achieve mixed-mode combustion, and the associated increase of combustion temperature promotes thermal NO formation. A portion of the formed $NO_x$ will be retained in the residuals, potentially affecting the next cycle. Therefore, the effects of $NO_x$ on mixed-mode combustion, especially on the end-gas auto-ignition, are investigated in the following. The numerical modeling approach validated earlier allows for such an investigation by activating or deactivating the $NO_x$ chemistry in the reaction model, which cannot be achieved in experimental studies.

Figures 11(*a*) and 11(*b*) show the pressure and apparent heat release traces calculated using reaction models with and without $NO_x$ chemistry. While the flame propagation stage before auto-ignition occurs is not significantly impacted by $NO_x$ chemistry, the end-gas behavior is significantly altered. In particular, no end-gas auto-ignition is observed when $NO_x$ chemistry is absent, implying that $NO_x$ plays a significant role in promoting auto-ignition. Such auto-ignition enhancement by $NO_x$ is attributed to enhanced chain branching due to the presence of nonnegligible $NO_x$-related radicals in the residual gas that can alter the reaction pathways during radical explosion and thereby modifying ignition delay. This is demonstrated in Fig. 11(*c*), showing that $Y_{CH_2O}$ is produced much earlier and faster with the presence of $NO_x$ than that obtained without $NO_x$ chemistry. Note that the in-cylinder NO mole fraction at the intake value closing point, averaging over all the cycles, is found to be $2.6 \times 10^{-4}$.

Figure 12 further shows the effects of $NO_x$ chemistry in 0D homogeneous auto-ignition and 1D flame propagation. As shown in

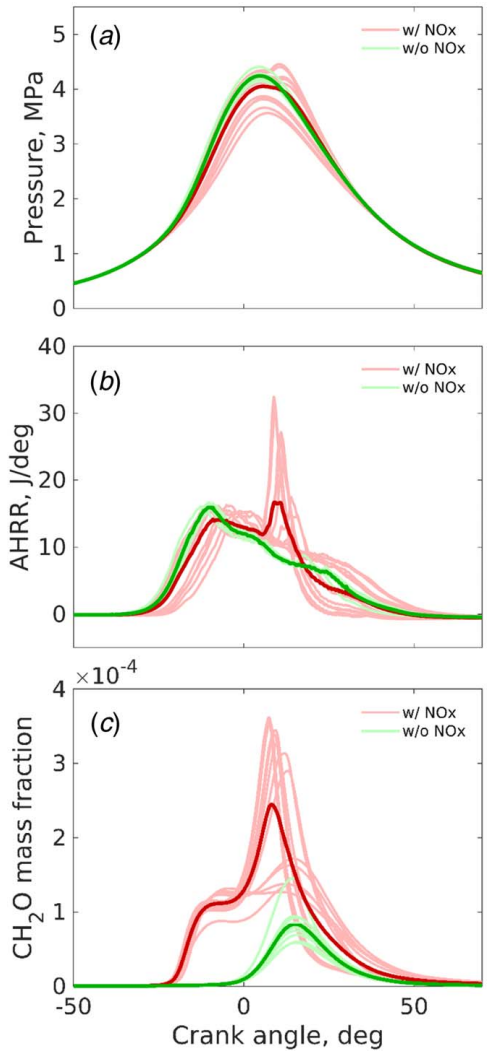



**Fig. 11 (*a*) Pressure, (*b*) apparent heat release rate, and (*c*) formaldehyde mass fraction profiles predicted by the skeletal model with and without $NO_x$ chemistry. Individual cycles are shown in light thin lines, while the mean values are shown in dark thick lines.**

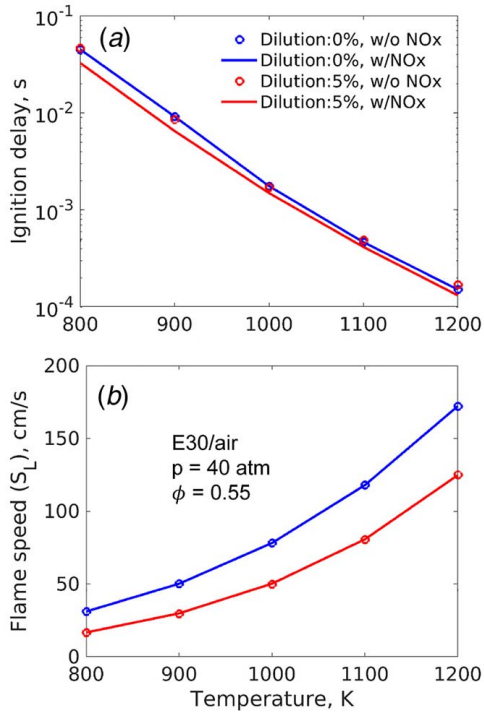



**Fig. 12 (*a*) Ignition delays and (*b*) flame speeds predicted by the skeletal reaction mechanism with and without $NO_x$ chemistry with residual gas fraction of 0% and 5%, respectively. Mole fractions of NO and $NO_2$ in the fresh mixture are $2.6 \times 10^{-4}$ and $1.7 \times 10^{-6}$, respectively, for residual gas fraction of 5%.**

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Fig. 12(*a*), the ignition delay calculated with and without $NO_x$ chemistry differs from each other when residual gas fraction (RGF) is not negligible, e.g., 5% (corresponding to the mean RGF for the present engine operation), in contrast to the case without any residual gas. $NO_x$, however, has only a very small effect on the laminar flame propagation regardless of the level of residual gas (Fig. 12(*b*)), as it is mainly controlled by back diffusion of sensible heat and important intermediates such as H and OH. It is therefore suggested that when RGF is nonnegligible, $NO_x$ chemistry has to be accounted for to accurately predict auto-ignition.

**Sensitivity to Fuel Properties.** Effects of physical and chemical properties of the E30 fuel, including heat of vaporization (HoV) and laminar flame speed ($S_L$), on mixed-mode combustion characteristics are then examined. Local sensitivity analysis was employed by perturbing the fuel properties by ±30% with respect to their nominal values.

Figures 13(*a*) and 13(*b*) show the pressure and heat release rate traces obtained from simulations using −30%HoV, HoV, and +30%HoV, respectively. A higher HoV is expected to have a negative impact on overall combustion intensity by reducing the overall in-cylinder temperature. It is seen that a perturbation on HoV has a negligible effect

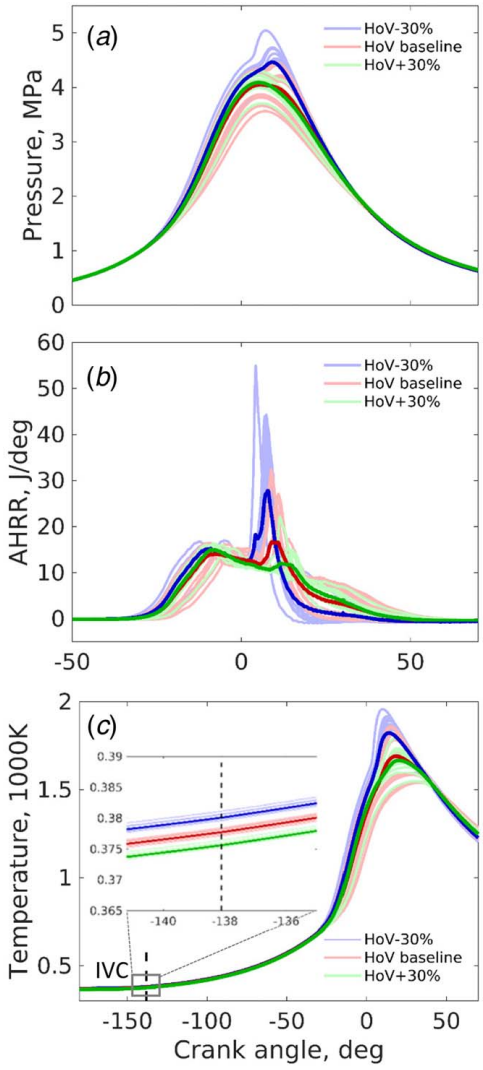



**Fig. 13 Sensitivity of predicted (*a*) pressure, (*b*) apparent heat release rate, and (c) cylinder-averaged temperature traces to heat of vaporization. Individual cycles are shown in light thin lines, while the mean values are shown in dark thick lines. IVC: intake value closing.**

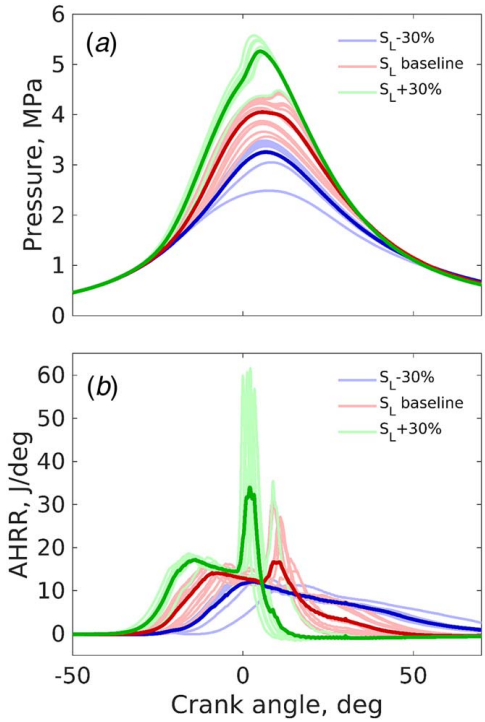



**Fig. 14 Sensitivity of predicted (*a*) pressure and (*b*) apparent heat release rate traces to laminar flame speed. Individual cycles are shown in light thin lines, while the mean values are shown in dark thick lines.**

on initial flame propagation. The auto-ignition-induced heat release rate is enhanced by a lower HoV, but suppressed with a higher HoV. The reason is that a higher HoV indicates the increased evaporation cooling, which induces a small reduction in in-cylinder temperature far before combustion occurs, as demonstrated in Fig. 13(*c*). Note that due to this charge cooling effect, changing HoV may also affect charged mass. However, for the perturbation range considered in this study, the effect of HoV on the in-cylinder equivalence ratio is found within 1% and therefore considered negligible.

In contrast, the effect of laminar flame speed has a much larger impact on mixed-mode combustion. This is shown in Fig. 14 for the pressure and heat release traces calculated using $-30\%S_L$, $S_L$, and $+30\%S_L$, respectively. Large $S_L$ not only advances the combustion phase but also increases the peak heat release rate during deflagrative flame propagation. As a result, subsequent auto-ignition is also advanced and intensified. In contrast, no auto-ignition is predicted for the lower $S_L$ and the overall heat release rate is much lower than the two larger $S_L$ values. Compared with the case with the nominal flame speed, the mean heat release rate at the first peak decreases by 23% for $-30\%S_L$ and increases by 22% for $+30\%S_L$.

Results shown earlier not only demonstrate the capability of the current modeling approach in capturing fuel property effects but also guide us to perform more detailed sensitivity analysis over a wider range of fuel properties. In particular, a coupled strategy of the neural network-based surrogate modeling approach, engine CFD, and global sensitivity analysis [32] could facilitate the understanding of the most influential fuel properties that enable mixed-mode combustion and lead to pathways for fuel-engine co-optimization. This topic will be addressed in the future work.

## Conclusions

A CFD model for lean, mixed-mode combustion in a DISI engine is developed in this work. Good agreement is observed between numerical results and experimental data, which demonstrates the capability of the developed CFD model in simultaneously

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characterizing deflagrative flame propagation and spontaneous auto-ignition for mixed-mode combustion. Moderate level of CCV is captured by simulation using an unsteady RANS approach. Instantaneous 3D flame structure reveals distinct combustion characteristics of deflagration-only cycles and mixed-mode cycles. Cycles with earlier flame propagation tend to produce mixed-mode combustion, since an advanced combustion phasing leads to the increased pressure and temperature which favor auto-ignition. In the mixed-mode cycles, isolated auto-ignition spots are observed which subsequently expand into the entire end-gas mixture. It is also seen that when both deflagration and auto-ignition are present, the deflagrative flame propagation is slightly suppressed by auto-ignition. The presence of auto-ignition is witnessed by the dramatically increased $CH_2O$ radical concentration. The positive correlation between occurrence of mixed-mode cycles and advanced CA50 is predicted in good agreement with experimental measurement.

The validated numerical model is then employed to investigate the effects of $NO_x$ chemistry and different fuel properties including HoV and laminar flame speed $S_L$. $NO_x$ chemistry is found to play an important role in promoting auto-ignition chemistry via retained residual gases, while the effect on flame propagation is minimal. Local sensitivity studies are performed to provide preliminary investigation of fuel property effects on mixed-mode combustion. Overall, flame propagation is not significantly modified with a perturbation in HoV, while a higher HoV reduces the auto-ignition tendency and peak heat release rate in the end-gas. An increase in the laminar flame speed significantly enhances the combustion phasing, for both deflagration and auto-ignition stages. A higher $S_L$ promotes flame propagation in terms of both combustion phasing and peak heat release. The enhanced flame propagation further enhances end-gas auto-ignition and raises the second peak in heat release rate since an advanced combustion phasing increases the compression heating of the end-gas. Across the parameter ranges studied here, the impact of $S_L$ is found much stronger than that of HoV on mixed-mode combustion.

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## Conflict of Interest

There are no conflicts of interest.

## Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request. The authors attest that all data for this study are included in the paper.

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